Project Synopsis

HCLTech Career Shaper Programme | IV Year Industry Project | 2026-27

Field	Details (Learner to complete)
Project Title	Time-Series Anomaly Detection in Financial Transactions
Project No.	58
Domain	Data Science / Python
Difficulty	Advanced
Duration	4 Weeks
Format	Individual Project
Learner Name	Shivam Verma
Roll Number	2300320130234
College / Campus	ABES Engineering College, Ghaziabad
Mentor Name	Natesan Rajan
Submission Date	29 October 2026

1. Problem Statement

Banks, payment providers, and financial analysts need to identify unusual transactions in large time-ordered transaction records. Rule-based checks may miss changing or previously unseen patterns, so this project explores unsupervised anomaly detection on transaction amounts, timestamps, and transaction categories to flag records for human review.

2. Proposed Solution & Approach

The project implements a Python-based financial transaction anomaly-detection prototype using Pandas for data preparation and Scikit-learn's Isolation Forest for unsupervised scoring. The workflow validates the CSV dataset, derives time-based and categorical features, and produces anomaly flags that can be inspected through a local Flask dashboard. Exploratory data analysis is provided through scripts and visualisations. The current dataset contains 500 records; the supplied anomaly labels and scores have unknown provenance, so the results are exploratory and require validation on trusted labelled data.

3. System Architecture

System block diagram: CSV transaction data → validation and preprocessing → feature engineering → Isolation Forest → anomaly results → local Flask dashboard and downloadable results. A formal Level 0 DFD should be confirmed/added if required by the mentor.

4. Technology Stack

Frontend / UI: HTML/CSS templates served by a local Flask dashboard
Backend / Logic: Python; Flask application and data-processing modules
Database: None; CSV files are used for input/output
AI/ML Framework: Scikit-learn (Isolation Forest)
Visualisation: Matplotlib; dashboard displays analysis/results
Hardware: Standard CPU-based computer; no special hardware required

5. Dataset Used

Dataset file name: dataset.csv
Number of rows: 500

Number of columns: 7
Key columns (list 5): timestamp, account_id, trans_amount, transaction_type, merchant_category
Any additional data collected: None reported; uses the project-folder dataset

6. Key Results & Metrics

Metric 1: Dataset size — 500 transactions across 7 columns
Metric 2: Automated tests — 24/24 Phase 2 tests passed in the recorded test run
Metric 3: Exploratory false-positive rate — approximately 0.604% at contamination 0.006 when compared with the supplied labels; label provenance is unknown, so this is not a validated performance metric
Business value achieved: Provides a reproducible prototype to prioritise unusual transactions for analyst review; operational effectiveness is not yet validated.

7. Challenges Faced & How Resolved

Challenge 1: Only 3 positive labels are present and label/score provenance is unknown → Resolution: treat evaluation as exploratory and document the need for verified ground-truth labels.
Challenge 2: The target false-positive rate of less than 0.1% has not been demonstrated → Resolution: tune and evaluate thresholds on a larger, independently labelled validation set before operational use.

8. References

1. Liu, F. T., Ting, K. M., and Zhou, Z.-H. (2008). Isolation Forest. IEEE International Conference on Data Mining. https://doi.org/10.1109/ICDM.2008.17
2. Scikit-learn Developers. IsolationForest documentation. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html
3. pandas Development Team. pandas documentation. https://pandas.pydata.org/docs/
4. Pallets Projects. Flask documentation. https://flask.palletsprojects.com/

9. Ethical Compliance Declaration

I confirm that:

☑ All dataset usage complies with the source licence (verify before signing).

☑ No personally identifiable information (PII) is stored without consent (verify before signing).

☑ AI tools used for code/writing are disclosed in the Appendix (add a disclosure appendix; AI assistance was used during project development/report preparation).

☑ This project complies with the DPDP Act 2023 where applicable (verify before signing).

Learner Signature: Shivam Verma Date: 29/10/2026